CHAPTER 09

# Reduce the impact of each journey

Environmental responsibility starts before choosing a fuel or a driving technique. Consider whether the journey is needed, which travel mode suits it and how to make a necessary drive safely and efficiently.

## In this chapter

**M09-L01** · Look beyond the exhaust

**M09-L02** · Use energy with foresight

**M09-L03** · Choose the journey as well as the car

**M09-L04** · Check where the vehicle can go

**M09-R** · Review and self-assessment

Answers and review guidance

Official references

This material is a learning aid to help you understand Swedish driving theory. It is not an official source and cannot replace official information from Trafikverket and Transportstyrelsen. Always consult their current guidance and the applicable rules.

M09-L01

# Look beyond the exhaust

**Your goal:** Compare climate, air-quality and noise effects without assuming that a vehicle has no environmental impact.

Transportstyrelsen distinguishes several effects. Greenhouse-gas emissions relate to the amount of travel, fuels, energy efficiency and electrification. Air pollution includes exhaust emissions and wear particles from brakes, tyres and the road. Noise is another effect, with tyre-road noise becoming important as speed rises. These are different measures: improving one does not prove that every environmental effect has disappeared. Climate and energy · Air quality · Noise

**Learning approach.** Ask what a comparison includes. Does an emissions figure describe a test result, the actual journey or a broader lifecycle? Are you comparing equal distances and occupancy? If the basis is missing, avoid a confident numerical conclusion. This lesson does not assign a universal emissions saving to a powertrain or declare a whole journey “zero impact.”

A plug-in hybrid combines electrical and combustion-engine operation. Transportstyrelsen changed the weighting used for new registrations in 2026 to better reflect the mix of those modes. The general lesson is to understand what a reported figure represents; it is not a promise about your next journey's consumption. Transportstyrelsen: plug-in-hybrid weighting

# Understand emissions

**Emissions explained · Fossil carbon and a cold catalyst.** Fossil petrol contains carbon stored for millions of years. Burning it combines that carbon with oxygen and releases $CO_2$ into the atmosphere. The added $CO_2$ strengthens the natural greenhouse effect and contributes to warming. Long-stored carbon: Naturvårdsverket · Climate effect: Trafikverket

A conventional petrol car's three-way catalyst reduces carbon monoxide, hydrocarbons and nitrogen oxides by chemical conversion. It needs sufficient heat to become effective, so it is least effective just after a cold start. Conversion products include water and $CO_2$: cleaning up these pollutants does not capture the fossil carbon or cancel its climate impact. No single warm-up time applies to every car. Transportstyrelsen: catalyst operation, page 24 · Trafikverket: cold emission controls

**Pollutant comparison.** Keep CO distinct from $CO_2$. The table summarises different mechanisms; one cleaner exhaust measure cannot stand for all of them.

| Substance | Why it matters |
|---|---|
| $CO_2$ · carbon dioxide | Added fossil $CO_2$ strengthens the greenhouse effect. |
| CO · carbon monoxide | Binds strongly to haemoglobin and interferes with the blood's oxygen transport. |
| NOx · nitrogen oxides | Can irritate airways, contribute to acidification and, with hydrocarbons and sunlight, form ground-level ozone. |
| Traffic particles | Exhaust and wear particles can enter and harm the respiratory system. |

Trafikverket: CO, NOx and particle mechanisms · Trafikverket: $CO_2$ and warming

# Compare powertrains and washing

**Powertrain and washing · Check the boundary.** A battery-electric car uses only electricity for propulsion and has no combustion tailpipe exhaust while driving. That does not remove impacts from vehicle/battery production, raw-material extraction or electricity supply. Tyre and road wear still matter. A plug-in hybrid combines electric and combustion operation and its traction battery can be charged externally. A conventional non-plug-in hybrid cannot be charged from an external electricity supply. Trafikverket: electric cars and hybrid types

Choose a purpose-built automatic or self-service wash that collects and treats its wastewater. Washing on an ordinary street or driveway can carry oil, metals and cleaning chemicals into ground or surface water. Treatment reduces the problem but does not remove every pollutant. Follow the facility's product instructions and local water-protection rules. Göteborg environmental authority: washing and treatment

**Apply the comparison.** A catalyst-equipped petrol car still releases fossil $CO_2$. An electric car avoids combustion exhaust on that drive, but needs a broader production-and-energy comparison. For either car, choose washing by how the dirty water is handled, rather than assuming a mild detergent makes street runoff harmless.

**Worked decision.** Two vehicles are advertised using different environmental measures. One figure describes exhaust emissions under a test procedure; another describes noise. The learner does not rank them as though both figures measure the same thing. They first identify the question being answered and the information still missing.

# Check your understanding

**Knowledge check.** (1) Why does fossil petrol warm the climate, even with a catalyst, and when is that catalyst least effective? (2) Match CO, NOx and particles to their effects. (3) Does an electric car have no environmental impact; which hybrid can charge externally? (4) Where should washing water go?

**Remember:** Name the effect and the basis of comparison before making the claim.

M09-L02

# Use energy with foresight

**Your goal:** Plan anticipatory driving that reduces unnecessary energy use while preserving safety.

Transportstyrelsen recommends correct tyre pressure, observing speed limits and removing unused roof racks or roof boxes. Driving technique and environmental awareness are part of driver education and assessment. Transportstyrelsen: economical driving

**Learning approach.** Look ahead and respond early to a developing queue rather than accelerating towards it and then braking sharply. Choose a suitable speed and preserve a gap that lets you adapt smoothly. This is not an instruction to delay necessary braking: safe control and traffic duties come first. If you need to stop for a person, do so regardless of what an efficiency display suggests. Traffic Regulation, chapter 3 §§2 and 14

Prepare the vehicle for the trip. Check the specified tyre pressure and remove unnecessary external carrying equipment when it is no longer needed. Combine these habits with realistic route and timing decisions. Avoid claiming a fixed percentage saving: load, road, weather, vehicle and technique all affect the outcome.

**Vehicle-specific boundary.** Gear selection and deceleration strategies vary. Do not use a fixed engine-speed rule for every car or assume every powertrain cuts fuel or recovers energy in the same way. Learn the actual vehicle's approved operation with your instructor. This book does not recommend coasting in neutral or switching the vehicle off while moving.

# Use energy with control

**Practical energy choices · Prepare and control the car.** Use a timer with an installed electric engine heater so it warms the engine when needed, without running unnecessarily long. Set it for the intended departure using the equipment instructions and conditions; there is no universal heating duration. Gagnef municipal energy advice: heater timer

Air conditioning consumes energy, but it can support comfort, alertness and clear windows. Avoid unnecessary cooling while keeping those safety benefits. Use the car's appropriate demisting controls; visibility takes priority over a saving. Cooling is not a remedy for sleep deprivation: stop and rest if drowsy. Revisit M07-L01 for seeing clearly and M07-L03 for attention. NZ Transport Agency: cooling, alertness and demisting

In a conventional manual car, choose a suitably high gear in which the engine runs smoothly without labouring. Change down if the load or gradient makes that gear unsuitable. Engine braking means slowing with the accelerator released while a suitable gear remains engaged; do not disengage the drive by pressing the clutch or selecting neutral for this technique. Use the brakes whenever safety requires. Actual engine control varies, so this definition does not promise zero fuel use or a universal RPM. Official vehicle agency: comfortable gear choice and in-gear deceleration

Unnecessary carried mass increases the demands on the vehicle. Unload heavy equipment you no longer need before future journeys, while retaining necessary safety and access equipment. Removing cargo is a different check from removing a roof box's air resistance. Swedish public energy advice: minimise weight

**Apply before departure.** Remove unused heavy equipment, schedule the heater for the trip, and keep effective demisting. On an uphill section, use the gear the car can comfortably handle; a high gear that strains the engine is not the intended economical technique.

# Look ahead and preserve room



The teal car approaches a queue on the right side of the road. Look ahead, retain room to stop and keep the crossing clear. The illustration does not specify a measured following distance.

**Worked decision.** You can see that a traffic queue is stationary ahead. You check behind and begin a controlled reduction in speed early, retaining enough room to stop. You do not accelerate simply to close the gap, and you do not roll into a crossing to preserve momentum.

# Check your understanding

**Knowledge check.** (1) May economy override a necessary stop? (2) How should you use an engine-heater timer and cooling? (3) What is a suitable high gear and engine braking in a manual car? (4) What should happen to unneeded heavy cargo?

**Remember:** Reduce avoidable demands on the car while keeping full control of the journey.

M09-L03

# Choose the journey as well as the car

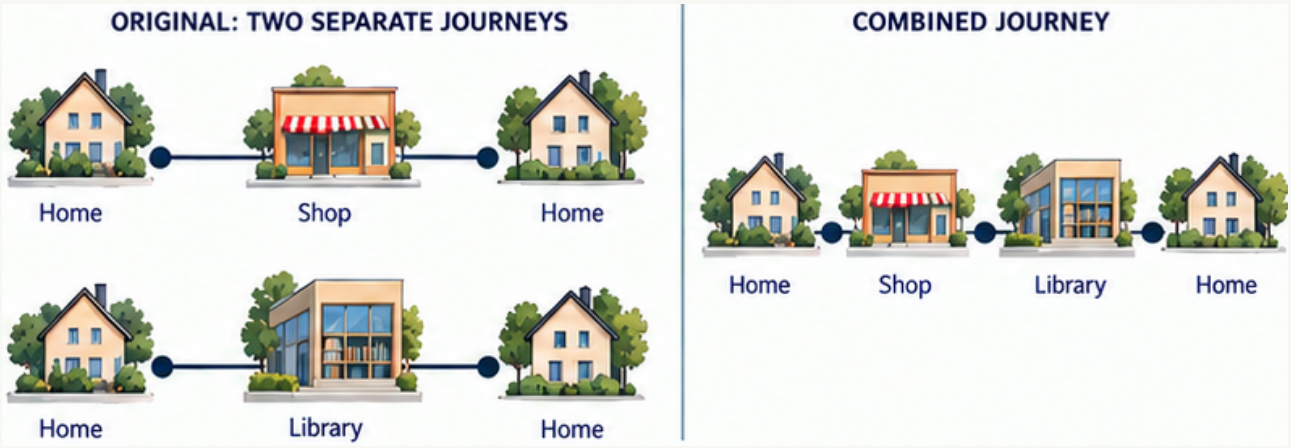
**Your goal:** Compare combining trips, walking, cycling, public transport and driving for a realistic personal journey.

The syllabus includes transport choices and the way personal circumstances influence them. A useful comparison recognises that people's access needs, schedules and available services differ. Category B syllabus, chapters 4–5

**Learning approach.** Start with the purpose: reaching work, collecting goods, accompanying a person or making a visit. Then list workable options. Could two errands be combined? Could a nearby trip be walked or cycled safely? Is public transport available for both directions? If you share a car journey, can everyone use the required protection and travel without creating a rushed schedule?

Do not replace one unrealistic assumption with another. A route may be short but inaccessible to a particular person. A service may run outward but not at the return time. A car may solve those needs, while a different choice suits the following day's journey. The exercise is to make the choice deliberately.

# Combine a practical journey




Separate journeys: home - shop - home, then home - library - home. Combined journey: home - shop - library - home. These are schematic options, not measured routes or a promised saving. Check practical needs and timing.

**Worked decision.** A learner normally makes two separate drives to the same district. On a day when both appointments can be combined without leaving anyone waiting unsafely, the learner plans one return trip. No exact fuel saving is claimed; the decision avoids a duplicated journey.

# Check your understanding

**Knowledge check.** Is there one transport mode that is automatically best for every journey and every person?

**Remember:** Good planning includes asking whether to drive, not only how to drive.

M09-L04

# Check where the vehicle can go

**Your goal:** Find official environmental-zone information and determine what to verify before entering.

**Requirements.** Municipalities can introduce environmental zones under the national framework. Class 1 concerns heavy buses and heavy goods vehicles; class 2 concerns passenger cars and light buses/goods vehicles. For class 2, spark-ignition vehicles need Euro 5 or Euro 6 and diesel vehicles need Euro 6. For light vehicles, class 3 ordinarily permits only electric, fuel-cell or qualifying gas vehicles; gas vehicles must meet the specified Euro 6 emissions criteria. A light plug-in hybrid is not admitted merely because it can sometimes run electrically. Exemptions exist for specified vehicles and uses. Transportstyrelsen: environmental zones · Traffic Regulation, chapter 4 §§22–24a

## Recognise environmental-zone starts and ends

Independent official examples. Match an ending to the same class at the start. These signs do not assert a current municipal boundary.




**E31 · Environmental zone**

Miljözon

The indicated environmental-zone rules apply. Check the zone class and whether your actual vehicle qualifies; the illustration does not establish a city's current boundary. Miljözon, class 1.

Transportstyrelsen · official description

# Continue the learning




**E31-2 · Environmental zone**

Miljözon

The indicated environmental-zone rules apply. Check the zone class and whether your actual vehicle qualifies; the illustration does not establish a city's current boundary. Miljözon, class 2.

Transportstyrelsen · official description




**E31-3 · Environmental zone**

Miljözon

The indicated environmental-zone rules apply. Check the zone class and whether your actual vehicle qualifies; the illustration does not establish a city's current boundary. Miljözon, class 3.

Transportstyrelsen · official description



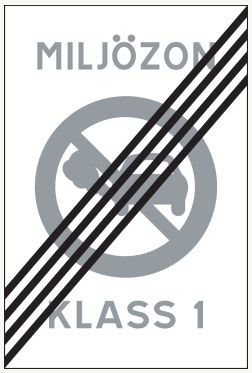

**E31-4 · Environmental zone**

Miljözon

The indicated environmental-zone rules apply. Check the zone class and whether your actual vehicle qualifies; the illustration does not establish a city's current boundary. Miljözon, classes 1 and 2.

Transportstyrelsen · official description

# Continue the learning

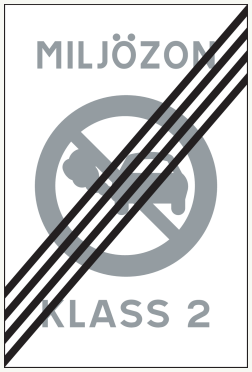



**E32 · End of environmental zone**

Slut på miljözon

The environmental zone ends. Class 1 environmental-zone sign crossed by diagonal lines.

Transportstyrelsen · official description

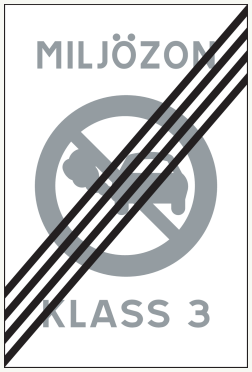



**E32-2 · End of environmental zone**

Slut på miljözon

The environmental zone ends. Class 2 environmental-zone sign crossed by diagonal lines.

Transportstyrelsen · official description




**E32-3 · End of environmental zone**

Slut på miljözon

The environmental zone ends. Class 3 environmental-zone sign crossed by diagonal lines.

Transportstyrelsen · official description

# Continue the learning




**E32-4 · End of environmental zone**

Slut på miljözon

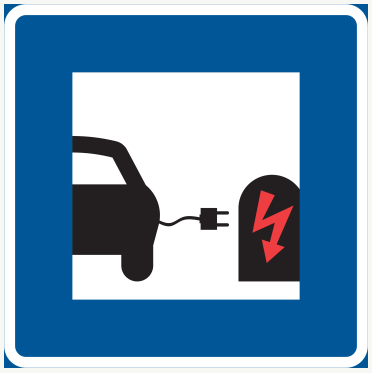
The environmental zone ends. Classes 1 and 2 environmental-zone sign crossed by diagonal lines.

Transportstyrelsen · official description

**Learning approach.** Check three layers: the national vehicle criteria, the current local restriction and your actual vehicle record. A national possibility to introduce a zone does not mean every city has introduced it. A manufacturer's marketing name does not prove the vehicle's recorded emissions class. Locate the applicable decision and signs for the actual route.

# Charging and parking are different

A charging facility and a parking qualification are different. Neither symbol gives permission to enter an environmental zone.



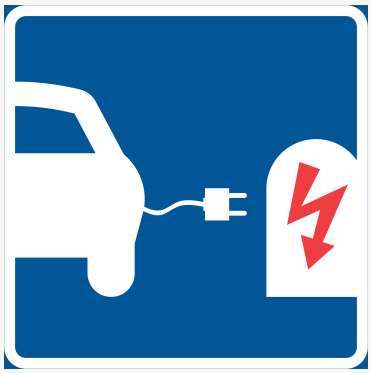
**H27 · Vehicle charging station**

Laddstation

This identifies a facility for externally charging an electrically powered vehicle. Car connected to a charging unit.

Transportstyrelsen · official description

# Continue the learning



**T24 · Parking for externally chargeable vehicles**

Laddplats

Below E19, only vehicles capable of external electric charging may park. The symbol alone does not say that charging must be in progress; check all posted conditions. Plug-in vehicle symbol.

Transportstyrelsen · official description

Transportstyrelsen's official traffic-regulation collection, STFS, provides local regulations. Check whether a decision is in force on the date of travel, and whether another decision has changed it. Do not rely on a remembered map or an announcement of a proposed zone. Official local traffic regulations

**Studded tyres · A separate local restriction.** Studs wear the road surface and create particles that can become airborne and enter the airways. Municipalities can restrict studded vehicles on particular roads or areas to address this problem. Check the current local decision in STFS, its dates, extent, exceptions and signs for your route. National permission to use studs in a season does not override a local prohibition. C44 denotes this restriction for motor vehicles other than moped class II; a supplementary panel can include that class too. Revisit the official C44 example in M07-L02. Trafikverket: stud-related particles · Transportstyrelsen: local restrictions, page 13 · Transportstyrelsen: C44

# Check access before leaving




Check the vehicle record and the current local decision before departure. The official E31 class-2 example is for recognition only; it does not establish a real boundary or permission for the illustrated car.

**Worked decision.** A driver has heard that a city plans a new zone. Before travelling, they verify the operative restriction and compare it with the car's recorded category. The route is chosen from current information, not from a headline remembered months earlier.

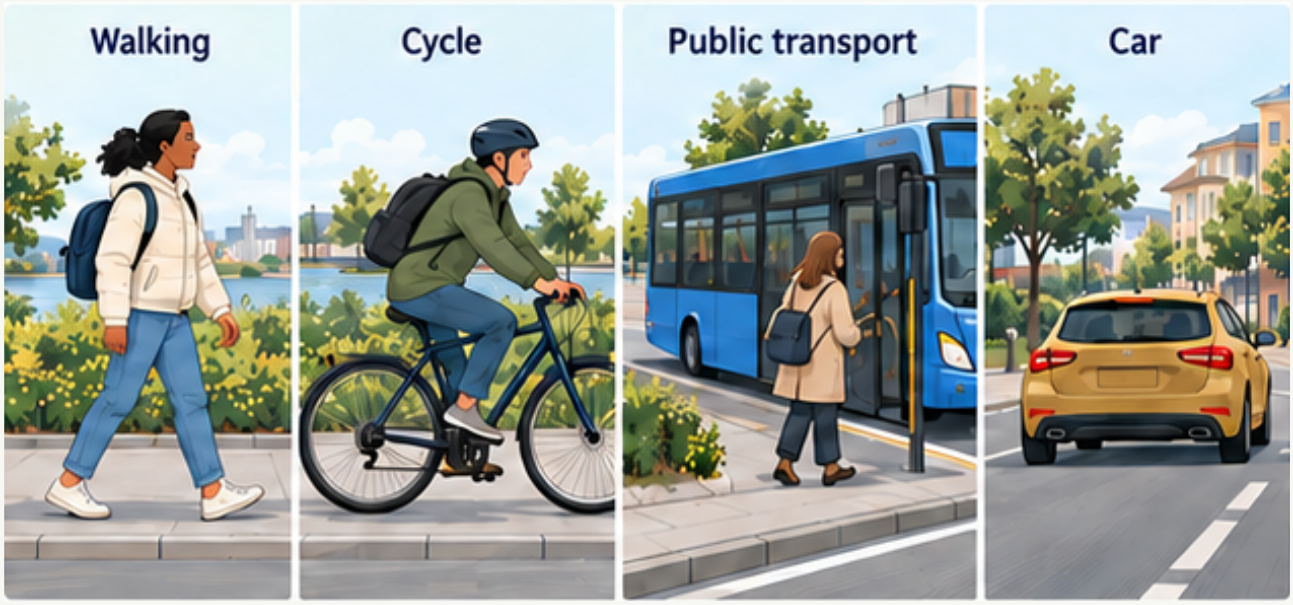
# Check your understanding

**Knowledge check.** (1) Does “hybrid” guarantee environmental-zone access? (2) Why can a municipality restrict studded tyres, and what must you check?

**Remember:** A rule's location and effective date are part of the rule.

# Review and self-assessment

**Scenario:** Plan a regular journey with an optional errand, two possible modes and a destination near a restricted area.




Walking, cycling, public transport and a car are options to compare against actual access needs, services and conditions. The pictures do not rank the modes or supply emissions figures.

Explain your choice without inventing savings figures.

# Keep a practice record

Record one assumption you made about environmental impact or access. Identify what evidence would change your decision and where you will verify it. Syllabus self-assessment, chapters 2, 4 and 5 §3

# Review after answering

M09-L01

**Answer:** (1) Long-stored carbon becomes $CO_2$, strengthening the greenhouse effect. A catalyst converts other pollutants and works poorly just after a cold start. (2) CO impairs blood oxygen transport; NOx irritates airways and contributes to acidification and ground-level ozone; particles harm airways. (3) No: production and electricity supply still matter. A plug-in hybrid can charge externally; a non-plug-in hybrid cannot. (4) Choose a purpose-built wash collecting and treating wastewater.

M09-L02

**Answer:** (1) No. (2) Heat when needed, avoiding unnecessary runtime; retain comfort, alertness and clear windows without needless cooling. (3) Use a gear allowing smooth running without strain. Engine braking releases the accelerator with a suitable gear engaged; brake as needed. No universal RPM or zero-fuel promise. (4) Unload it before future trips; retain necessary equipment.

M09-L03

**Answer:** No. Compare the actual needs, conditions and available options, then avoid unnecessary travel where practical.

M09-L04

**Answer:** (1) No. Check zone class, vehicle criteria, exemptions and current decision. (2) Studs wear roads and create airborne particles. Check the local decision, route, dates, exceptions and signs; national permitted season does not override a local ban.

M09-R

**Review guide:** Compare realistic alternatives, avoid unnecessary distance, identify restrictions and keep safety ahead of efficiency during any drive.

# Official references

Original references checked 7 October 2026; G9 additions checked 13 October 2026. Sign captions and artwork use the official asset register. Consult current guidance and applicable rules.

### Climate and energy

https://www.transportstyrelsen.se/sv/vagtrafik/miljo/klimat-och-energi/

### Air quality

https://www.transportstyrelsen.se/sv/vagtrafik/miljo/luftkvaliet-i-tatorter/

### Noise

https://www.transportstyrelsen.se/sv/vagtrafik/miljo/buller/

### Transportstyrelsen: plug-in-hybrid weighting

https://www.transportstyrelsen.se/sv/om-oss/pressrum/nyhetsarkiv/2026/forandrad-viktningsfaktor-for-nya-laddhybrider/

### Long-stored carbon: Naturvårdsverket

https://www.naturvardsverket.se/amnesomraden/klimatforandringar/darfor-blir-det-varmare/kolets-kretslopp-rubbas/

# Official references

Original references checked 7 October 2026; G9 additions checked 13 October 2026. Sign captions and artwork use the official asset register. Consult current guidance and applicable rules.

### Climate effect: Trafikverket

https://www.trafikverket.se/resa-och-trafik/trafiksakerhet/sakerhet-pa-vag/hastighetsgranser-pa-vag/hastighet-och-hallbarhet/

### Transportstyrelsen: catalyst operation, page 24

https://www.transportstyrelsen.se/globalassets/global/publikationer-och-rapporter/rapporter/vagtrafik/atgarder-mot-manipulering-av-viss-fordonsutrustning.pdf

### Trafikverket: cold emission controls

https://bransch.trafikverket.se/for-dig-i-branschen/miljo---for-dig-i-branschen/luftkvalitet/vagtrafikens-luftfororeningar/

### Trafikverket: electric cars and hybrid types

https://www.trafikverket.se/om-oss/skydda-klimat-miljo-och-halsa-i-vart-arbete/elfordon--sa-funkar-det/

### Göteborg environmental authority: washing and treatment

https://goteborg.se/wps/portal/start/bygga-bo-och-leva-hallbart/leva-hallbart/smarta-resor/tvatta-inte-bilen-pa-gatan

# Official references

Original references checked 7 October 2026; G9 additions checked 13 October 2026. Sign captions and artwork use the official asset register. Consult current guidance and applicable rules.

### Transportstyrelsen: economical driving

https://www.transportstyrelsen.se/sv/vagtrafik/miljo/klimat-och-energi/sparsam-korning/

### Traffic Regulation, chapter 3 §§2 and 14

https://www.riksdagen.se/sv/dokument-och-lagar/dokument/svensk-forfattningssamling/trafikforordning-19981276_sfs-1998-1276/

### Gagnef municipal energy advice: heater timer

https://www.gagnef.se/bygga-bo-och-miljo/boende-och-miljo-2/energi-och-klimatradgivning/

### NZ Transport Agency: cooling, alertness and demisting

https://nzta.govt.nz/vehicles/choosing-the-right-vehicle/features-that-affect-the-environment

### Official vehicle agency: comfortable gear choice and in-gear deceleration

https://www.vehicle-certification-agency.gov.uk/fuel-consumption-co2/fuel-consumption-guide/fuel-efficient-driving-tips/

# Official references

Original references checked 7 October 2026; G9 additions checked 13 October 2026. Sign captions and artwork use the official asset register. Consult current guidance and applicable rules.

### Swedish public energy advice: minimise weight

https://energiochklimatradgivningen.se/hushall/resahallbart/sparsamkorning.685.html

### Category B syllabus, chapters 4–5

https://www.transportstyrelsen.se/TSFS/TSFS%202011_20.pdf

### Transportstyrelsen: environmental zones

https://www.transportstyrelsen.se/sv/vagtrafik/miljo/miljozoner/

### Official local traffic regulations

https://rdt.transportstyrelsen.se/rdt/Default.aspx?UseStfs=1

### Transportstyrelsen: local restrictions, page 13

https://www.transportstyrelsen.se/globalassets/global/publikationer-och-rapporter/rapporter/vagtrafik/utredning-om-nar-det-ska-vara-tillatet-att-anvanda-dubbdack.pdf

# Official references

Original references checked 7 October 2026; G9 additions checked 13 October 2026. Sign captions and artwork use the official asset register. Consult current guidance and applicable rules.

## Transportstyrelsen: C44

https://www.transportstyrelsen.se/sv/vagtrafik/trafikregler-och-vagmarken/vagmarken/forbudsmarken/forbud-mot-trafik-med-annat-motordrivet-fordon-med-dubbdack-an-moped-klass-ii/

Road-sign illustrations: Transportstyrelsen. Each reference example links to its official description. Swedish names are official; English labels and explanations are editorial translations. Road scenes are simplified illustrations, not to scale.